

CENTAUR CONSULTANTS

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TURBINE NOZZLE PARTS DRAWING

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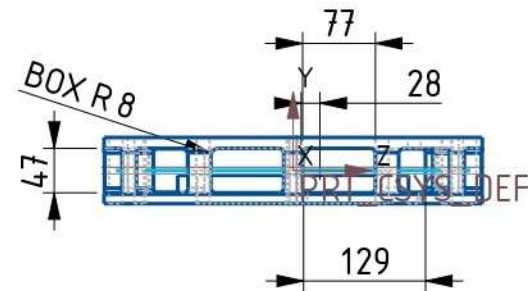
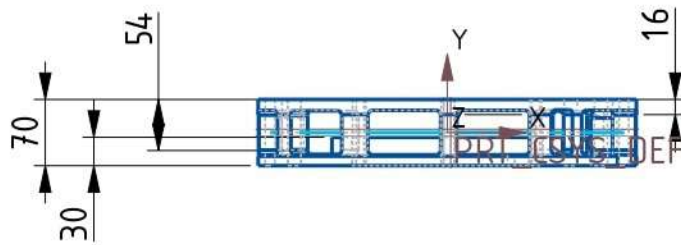
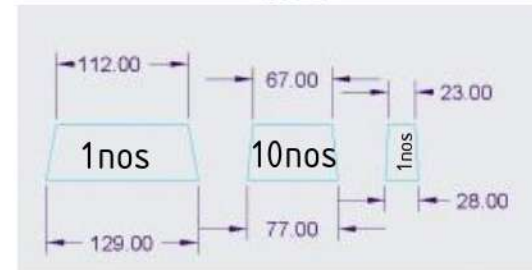
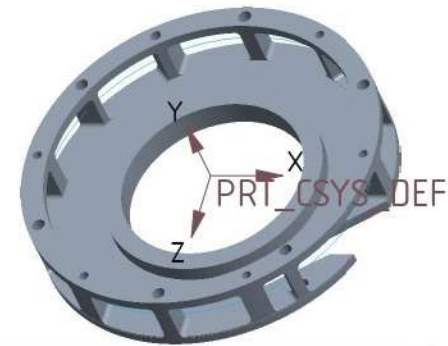
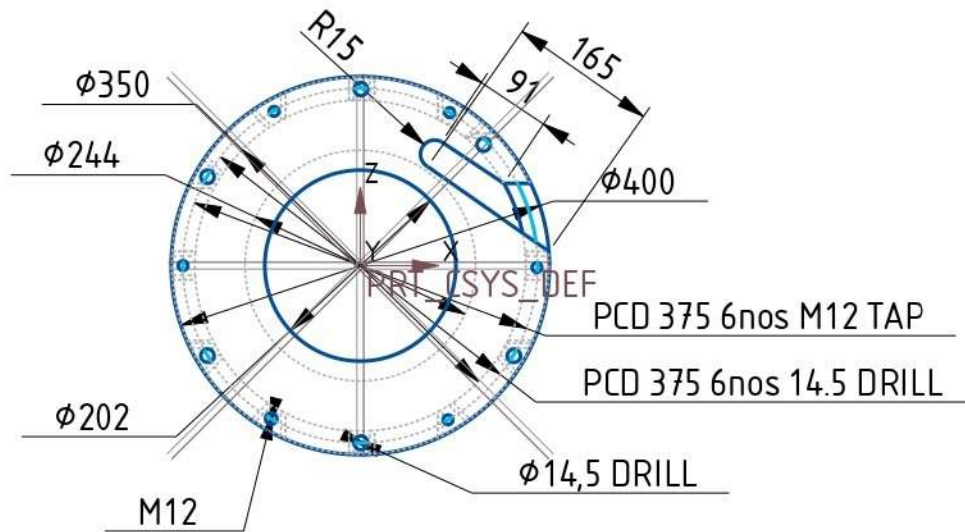
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If there is any clarification regarding the drawing?

Please contact Name : AATHEEF AKRAM

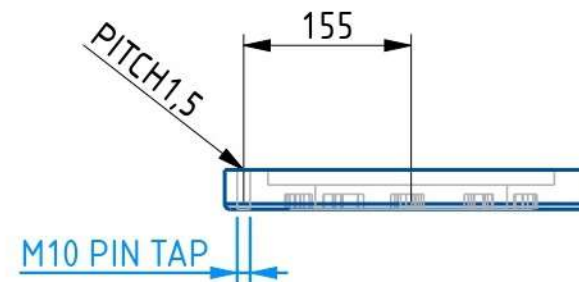
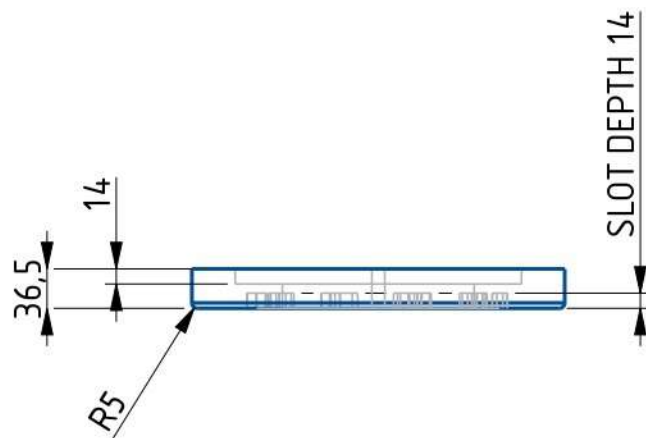
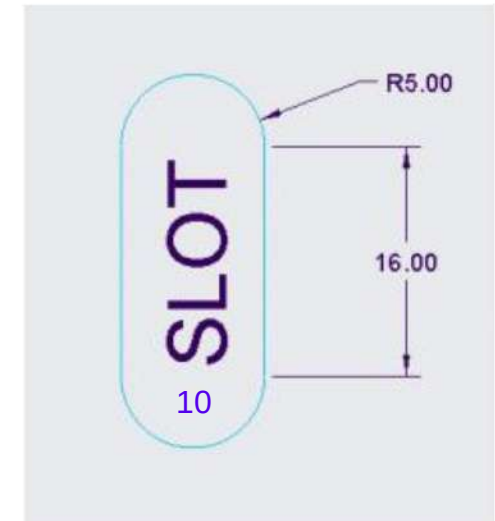
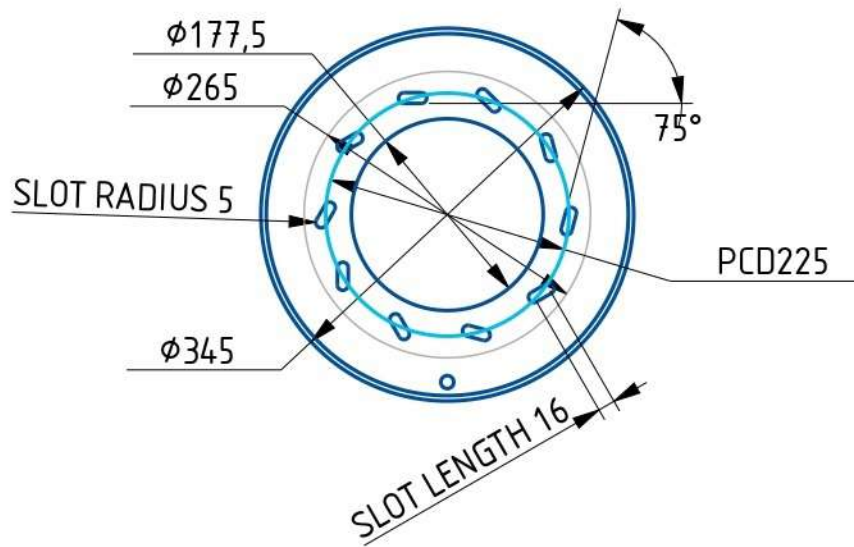
Phone No : 9790846768



NOTES :

1. ALL DIMENSION ARE IN mm
2. SHARP EDGES, BURR SHOULD BE REMOVED
3. TOLERANCE UNSPECIFIED : $\pm 0.1\text{mm}$

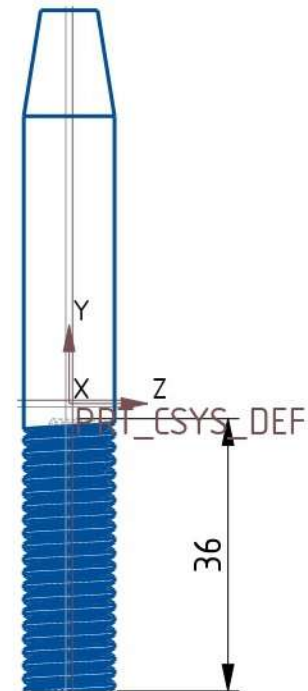
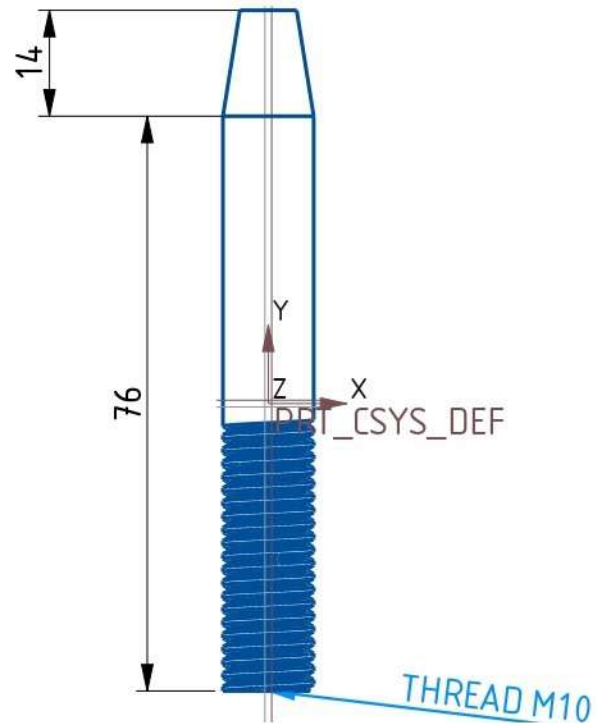
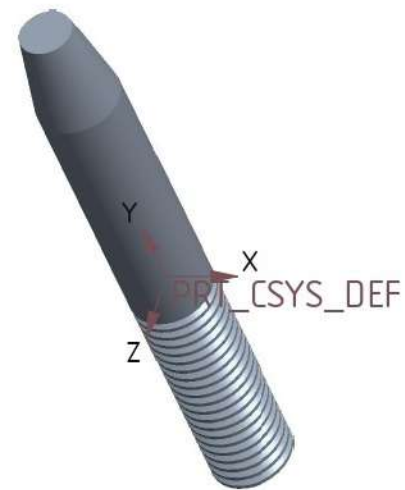
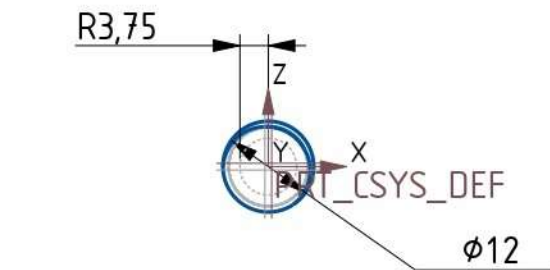
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	DRG TITLE : TOP PART
	PART NO :
	DRG NO : 1
SCALE	1:1



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1. ALL DIMENSION ARE IN mm
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	CENTAUR CONSULTANTS	
	DRG TITLE : MIDDLE PART	
	PART NO :	
	DRG NO : 2	
SCALE	1:1	



NOTES :

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CENTAUR CONSULTANTS

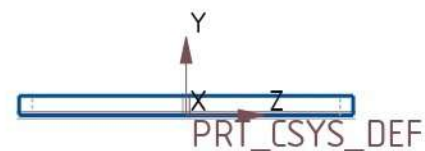
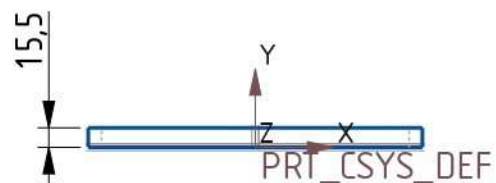
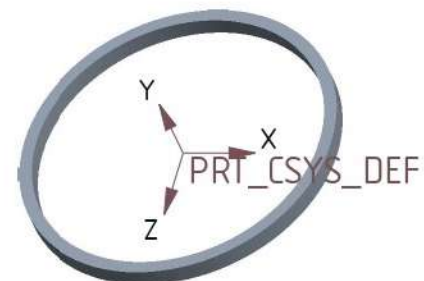
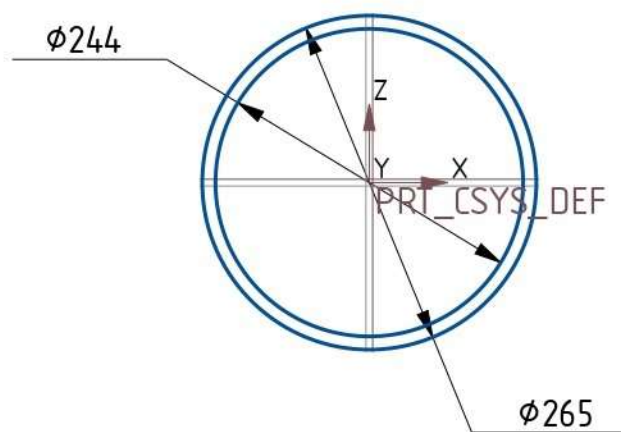
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PART NO :

DRG NO : 3

SCALE

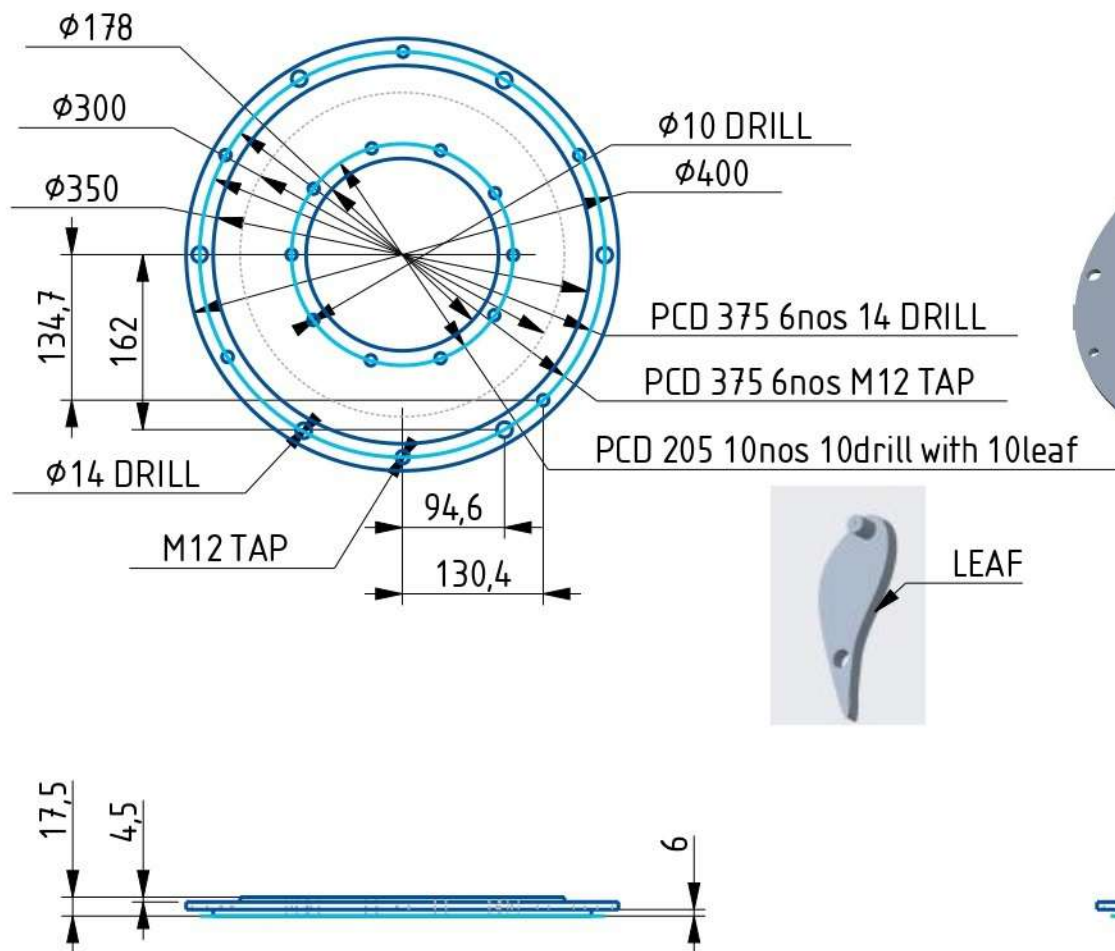
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NOTES :

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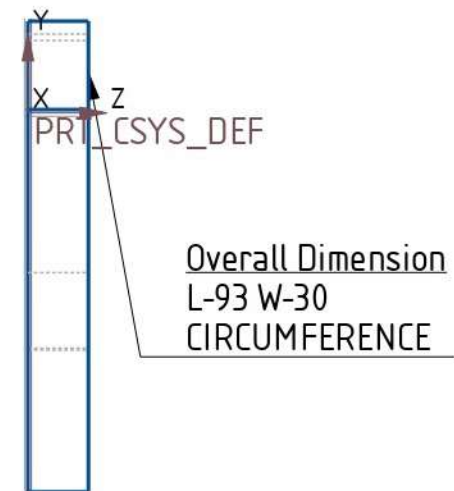
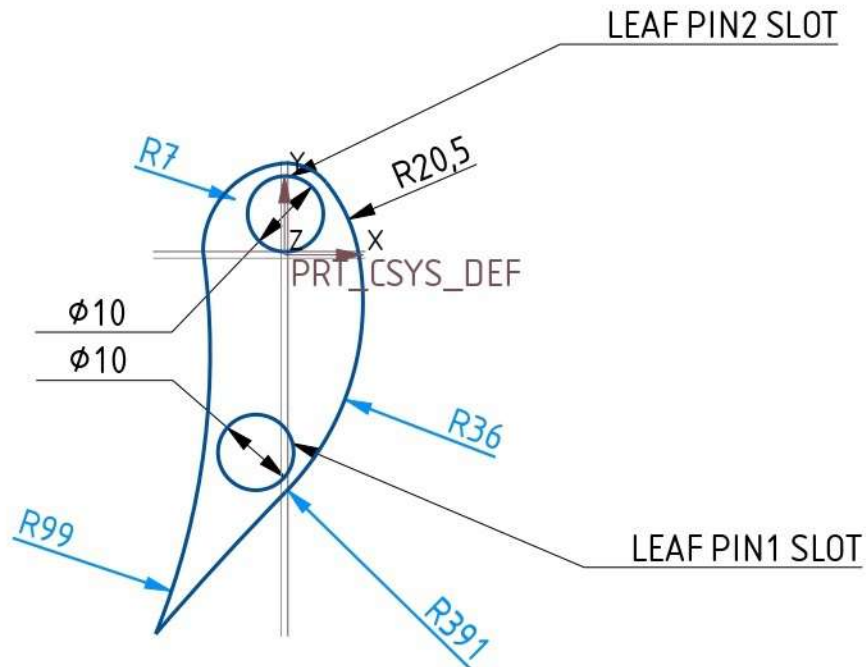
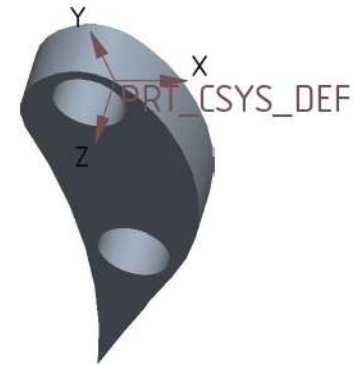
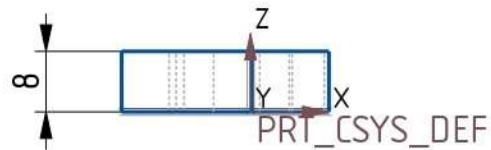
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SCALE		
1:1		



NOTES :

1. ALL DIMENSION ARE IN mm
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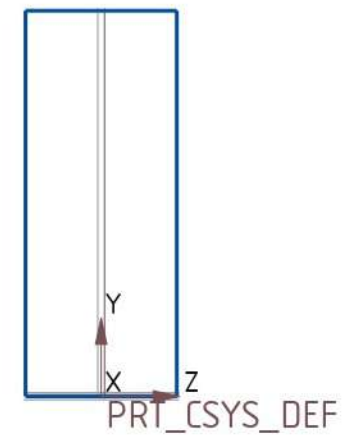
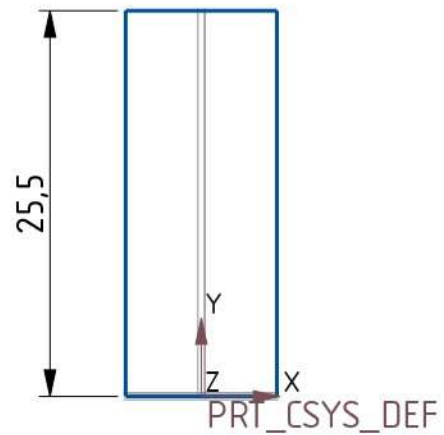
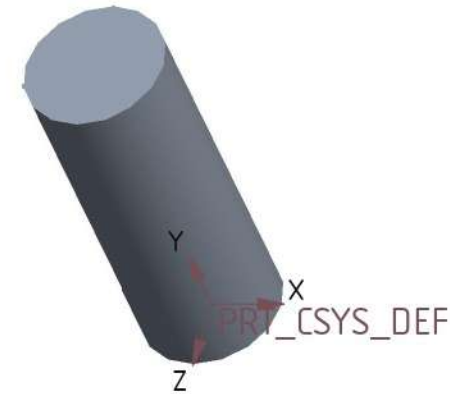
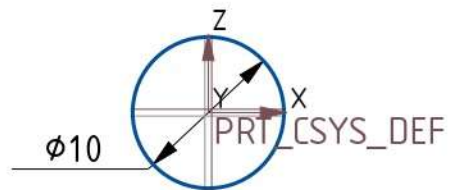
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	DRG TITLE : BOTTOM PART	
	PART NO :	
	DRG NO : 5	
SCALE		
1:1		



NOTES :

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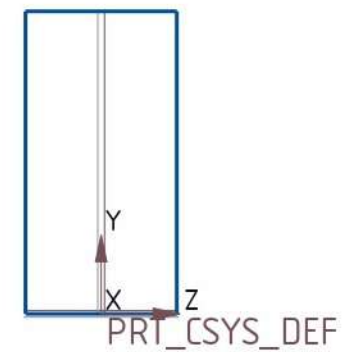
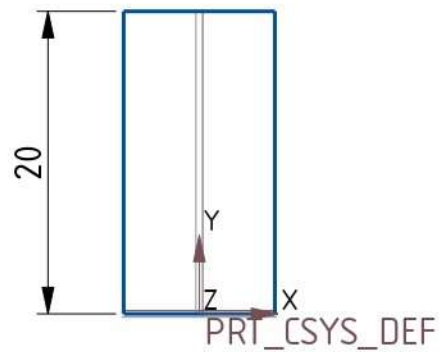
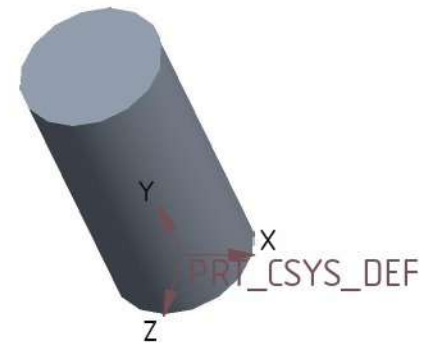
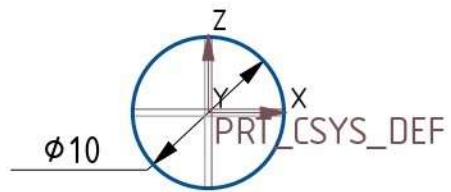
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	DRG TITLE : LEAF PART	
	PART NO :	
	DRG NO : 6	
SCALE	1:1	



NOTES :

1. ALL DIMENSION ARE IN mm
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3. TOLERANCE UNSPECIFIED : $\pm 0.1\text{mm}$

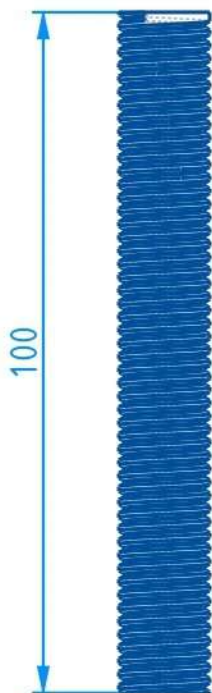
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	DRG TITLE : LEAF PIN 1	
	PART NO :	
	DRG NO : 7	
SCALE		
1:1		



NOTES :

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	CENTAUR CONSULTANTS	
	DRG TITLE : LEAF PIN 2	
	PART NO :	
	DRG NO : 8	
SCALE	1:1	



NOTES :

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3. TOLERANCE UNSPECIFIED : $\pm 0.1\text{mm}$



CENTAUR CONSULTANTS

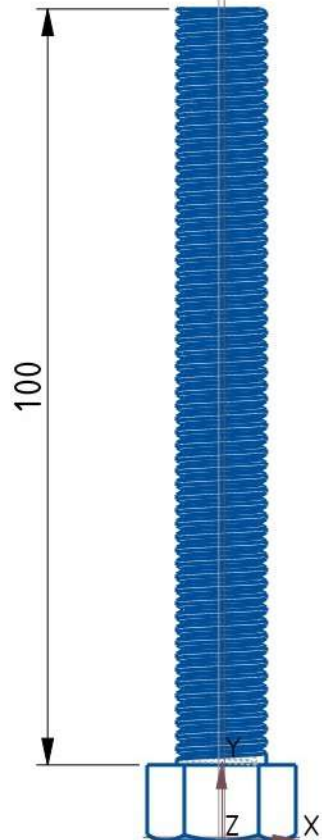
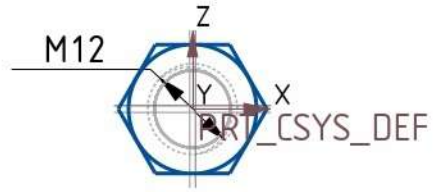
DRG TITLE : STUD PART

PART NO :

DRG NO : 9

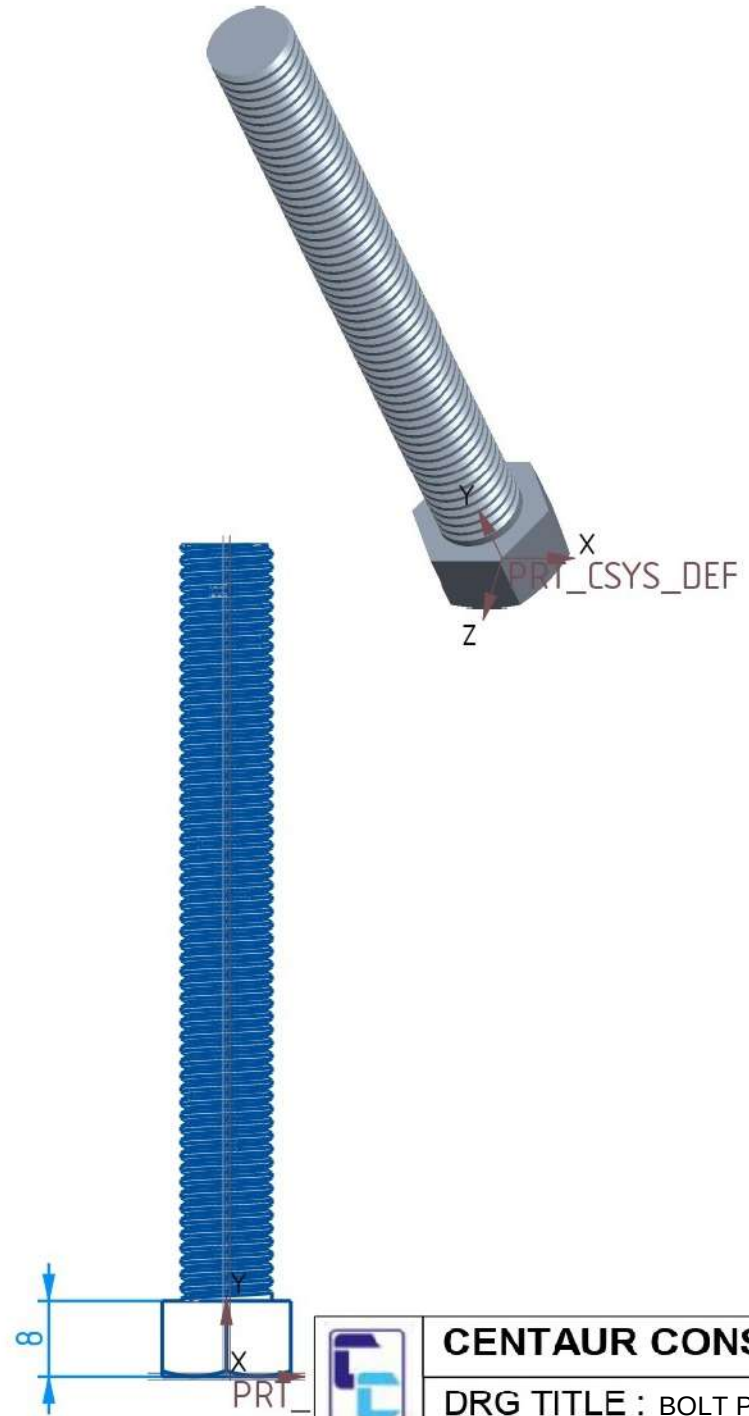
SCALE

1:1

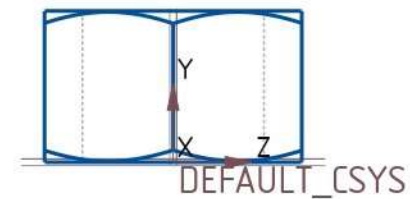
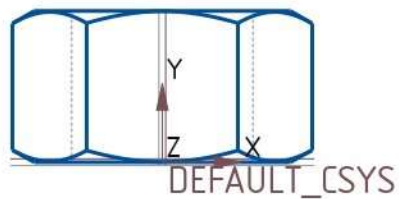
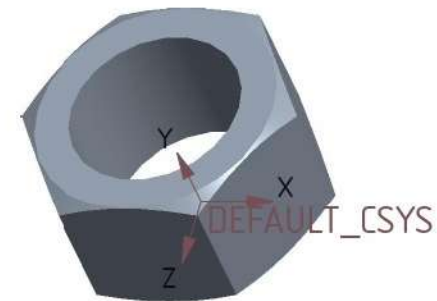
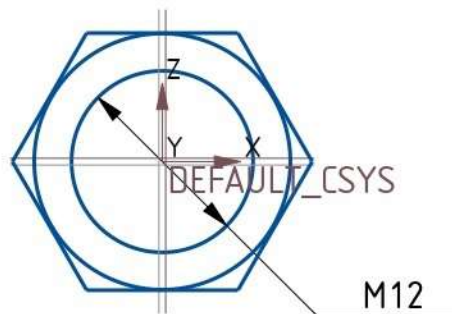


- NOTES :**
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10



	CENTAUR CONSULTANTS	
	DRG TITLE : BOLT PART	
	PART NO :	
	DRG NO : 10	
SCALE	1:1	

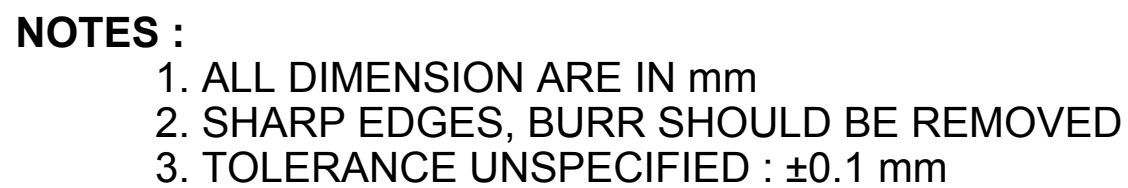


NOTES :

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3. TOLERANCE UNSPECIFIED : $\pm 0.1\text{mm}$

	CENTAUR CONSULTANTS	
	DRG TITLE : NUT PART	
	PART NO :	
	DRG NO : 11	
SCALE		
1:1		

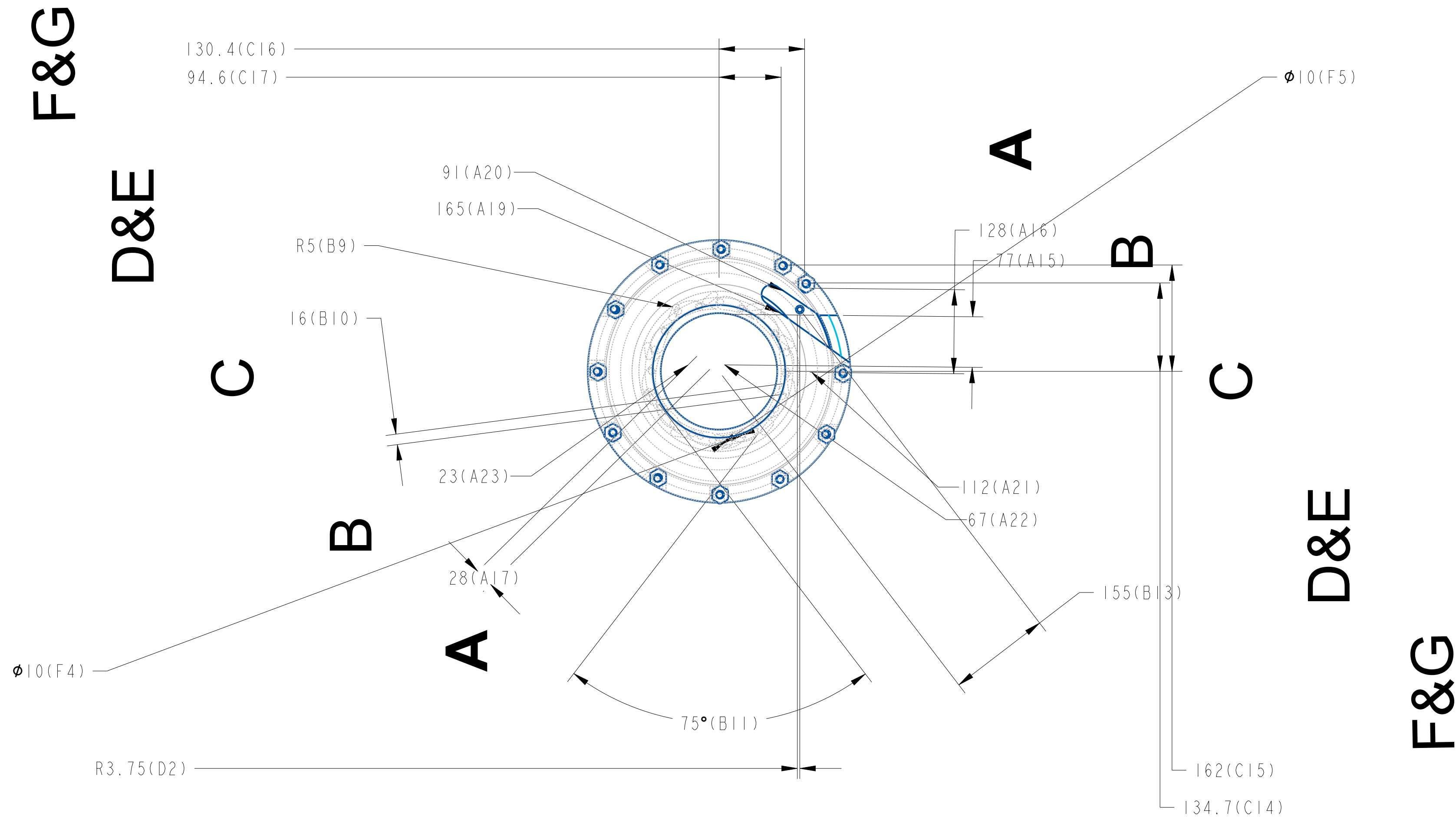
12



	CENTAUR CONSULTANTS
	DRG TITLE : TURBINE ASSEMBLY FRONT VIEW
SCALE	PART NO :
1:1	DRG NO : 12

TURBINE ASSEMBLY

TOP VIEW



NOTES :

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ELABORATION OF DIMENSION IS IN TABLE			
A	TOP PART	C&E	MIDDLE PART PIN & CARBON
B	MIDDLE PART	F&G	BOTTOM PART LEAF &
C	BOTTOM PART		GENERAL PARTS

	CENTAUR CONSULTANTS	
	DRG TITLE : TURBINE ASSEMBLY TOP VIEW	
	PART NO :	
	DRG NO : 13	
SCALE	1:1	

ASSEMBLY DATA TABLE

<u>NAME</u>	<u>NO</u>	<u>Size</u> <u>(mm)</u>	<u>DESCRIPTION</u>
TOP PART (A)			Top part of the turbine nozzle
A	1	∅ 400	Over all outer Diameter of the part
A	2	∅ 350	inner diameter of the part
A	3	PCD 375	6nos M12 tap in the part
A	4	PCD 375	6nos 14.5 drill in the part
A	5	∅ 244	Outer diameter of under cut
A	6	∅ 202	Over all Inner diameter
A	7	∅ 14.5	Drill size
A	8	M12	M12 Tap thread
A	9	R15	Radius of the U-cut
A	10	70	Over all thickness of the part
A	11	54	Depth of the under cut
A	12	30	Thickness of the plate than the under cut

A	13	16	Shape length for surface finish
A	14	R8	Fillet for the boxes
A	15	77	10nos box in the side of the part length
A	16	129	1nos biggest box of the part length
A	17	28	1nos smallest box of the part-length
A	18	47	Width of the all boxes
A	19	165	Long length of the U-cut
A	20	91	Short length of the U-cut
A	21	112	Tapper length of 129 box
A	22	67	Tapper length of 77 box 10nos
A	23	23	Tapper length of 28 box
MIDDLE PART (B)			Middle part of the turbine nozzle
B	1	ø345	Over all outer diameter
B	2	ø265	Under cut inner diameter
B	3	PCD 225	PCD of the 10nos slots in the part
B	4	ø117.5	Over all inner diameter
B	5	36.5	Overall thickness of the part
B	6	14	Under-cut depth
B	7	R5	Round in the part OD

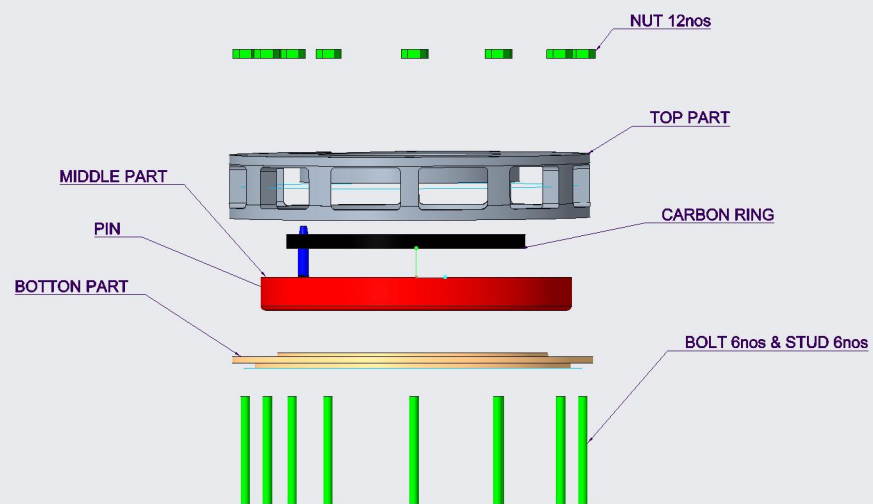
B	8	14	Slot depth
B	9	R5	Slot radius
B	10	16	Slot length
B	11	75°	Angle of the slot between them
B	12	1.5	Pitch of the thread for pin
B	13	155	Distance from the centre to pin
BOTTOM PART(C)			Bottom part of the turbine nozzle
C	1	ø 400	Over all outer diameter of the part
C	2	PCD 375	PCD of 6nos 14 drill
C	3	PCD 375	PCD of 6nos M12 tap
C	4	ø350	Outer diameter of bottom
C	5	ø300	Outer diameter of top
C	6	ø178	Overall inner diameter
C	7	PCD 205	PCD of 10nos 10 drill for 10 leaf's
C	8	ø14	Drill inner diameter in PCD 375
C	9	M12	M12 thread in PCD 375
C	10	ø10	Drill inner diameter in PCD 205
C	11	17.5	Over all thickness of the part
C	12	4.5	Thickness from 300 outer diameter

C	13	6	Thickness from 350 outer diameter
C	14	134.7	Unique hole M12 height from the centre
C	15	162	Unique hole 14drill height from the centre
C	16	130.4	Unique hole M12 length from the centre
C	17	94.6	Unique hole 14drill length from the centre
MIDDLE PART PIN & CARBON (D&E)			The pin and the carbon ring will fit inside The middle part
D			PIN PART
D	1	ø12	Overall Pin outer diameter
D	2	R3.75	Tapper outer diameter of the pin
D	3	14	Tapper height
D	4	76	Height of the pin without tapper height
D	5	36	Thread length from the bottom
D	6	M10	Thread type
E			CARBON RING
F	1	ø265	Over all outer diameter
F	2	ø244	Over all inner diameter
F	3	15.5	Thickness of the ring
BOTTOM PART LEAF&			Leaf part and other general parts used in the

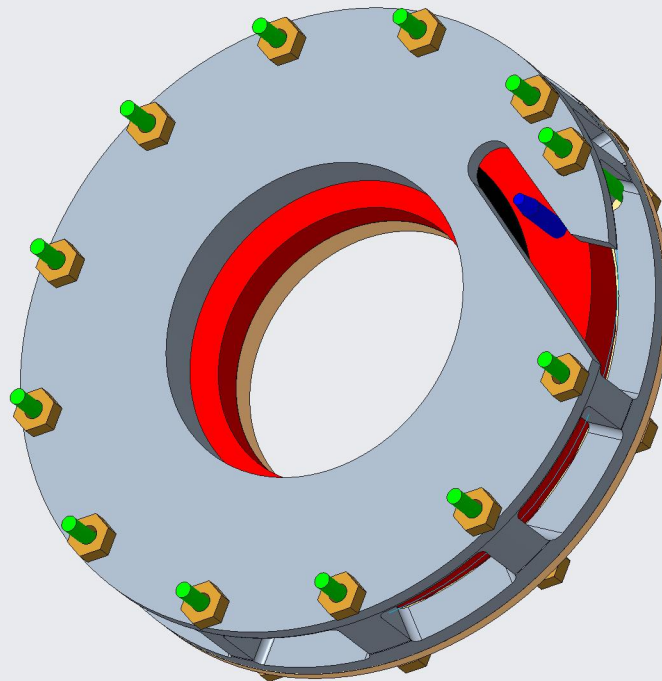
GENERAL PARTS(F&G)			assembly
F			LEAF PART
F	1	8	Over all thickness of the leaf
F	2	R7	Radius of left arc of leaf
F	3	∅ 10(Pin 2)	Leaf Pin 2 inner diameter
F	4	∅10	Pin 2 slot Hole drill
F	5	R20.5	Radius of right arc of leaf
F	6	∅10	Pin 1 slot Hole drill
F	7	R36	Belly radius of the leaf
F	8	R391	Tail radius of the leaf
F	9	R99	Straight radius of the leaf
F	10	∅ 10(Pin 1)	Leaf Pin 1 inner diameter
F	11	25.5	Height of pin 1
F	12	20	Height of pin 2
G			GENERAL PARTS (standard bolt, stud & nut)
G	1	M12	Stud, Bolt & nut size
G	2	100	Stud length
G	3	120	Bolt length

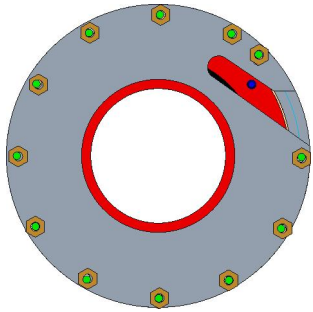
MATERIAL DATA TABLE

<u>S.No.</u>	<u>NAME</u>	<u>SYMBOL</u>	<u>MATERIAL</u>
1	TOP PART	A	SS 316 L
2	MIDDLE PART	B	SS 316 L
3	BOTTOM PART	C	SS 316 L
4	MIDDLE PART PIN	D	SS 316 L
5	MIDDLE PART CARBON RING	E	GRAPHITE TEFLON
6	LEAF	F	SS 316 L
7	LEAF PIN1	F	SS 316 L
8	LEAF PIN 2	F	SS 316 L
9	STUD	G	SS 316 L
10	BOLT	G	SS 316 L
11	NUT	G	BRASS

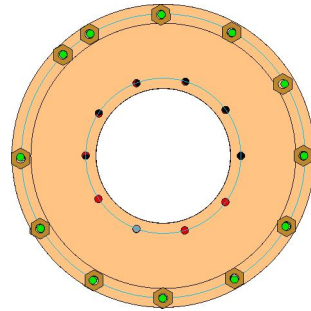


ASSEMBLY MERGE VIEW

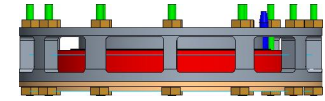




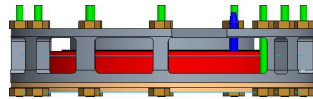
TOP ORIENTATION



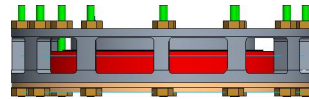
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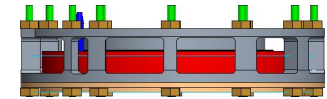
FRONT ORIENTATION



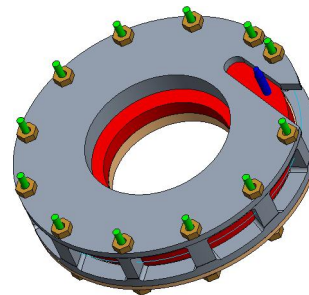
RIGHT ORIENTATION



LEFT ORIENTATION



BACK ORIENTATION



STANDARD ORIENTATION